

AI Foundations for Senior Engineers (Week 1-2)

Program Context

This program is designed for **senior Node.js engineers (10+ years of experience)** who already understand APIs, NLP terminology, and how to invoke LLMs, but lack clarity on: - Core AI fundamentals - Problem identification and framing - Deciding *when* and *how* AI should be used - Explaining AI decisions to management and interviewers

The first **two weeks** are intentionally focused on **mental models, clarity, and AI thinking**, not heavy mathematics or tooling.

Program Structure (Weeks 1-2)

- **Duration:** 2 weeks (10 working days)
- **Daily Time:** 2 hours/day
- **1 hour:** Conceptual teaching
- **1 hour:** Progressive discussion / exercises

Primary Objective:

Transition participants from *"I can call an AI API"* to *"I understand what kind of AI problem this is and why."*

WEEK 1 — Understanding What AI Actually Is

Day 1 — The AI Landscape (Clarity Over Hype)

Teaching (1 Hour)

- What AI really means (vs hype)
- AI vs Machine Learning vs Deep Learning vs LLMs
- Why LLMs are not equal to AI
- What problems AI is good at vs bad at

Key Mental Model:

AI is pattern learning from data, not magic or reasoning.

Progressive Session (1 Hour)

- Group discussion: Where have you already used AI unknowingly?

- Exercise: Identify features that are AI vs non-AI problems

Learning Resources

- Andrew Ng – *AI for Everyone*
<https://www.coursera.org/learn/ai-for-everyone>
 - StatQuest – *Machine Learning in 5 Minutes*
<https://www.youtube.com/watch?v=ukzFI9rgwfU>
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Day 2 — Machine Learning Intuition (No Math Focus)

Teaching

- What Machine Learning actually learns
- Inputs → Model → Outputs
- What training means conceptually
- Supervised vs Unsupervised vs Reinforcement Learning

Progressive Session

- Classify real business problems into ML categories
- Identify problems that should not use ML

Learning Resources

- Google Machine Learning Crash Course (Intro)
<https://developers.google.com/machine-learning/crash-course>
 - StatQuest – *What is Machine Learning?*
https://www.youtube.com/watch?v=Gv9_4yMHfHl
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Day 3 — Classification vs Regression (The Core of ML)

Teaching

- Why most ML problems reduce to classification or regression
- Classification: categories and decisions
- Regression: numeric prediction
- Mapping LLM use cases back to classical ML thinking

Progressive Session

- Convert vague requirements into classification or regression problems

Learning Resources

- StatQuest – *Classification vs Regression*
<https://www.youtube.com/watch?v=8jazNUpO3lQ>
 - Google ML Crash Course – Problem Framing
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Day 4 — Neural Networks (Conceptual Understanding)

Teaching

- Why neural networks exist
- Neurons as weighted decision units
- Layers and representations
- Why deep learning works well for text, images, and audio

Progressive Session

- Visual walkthroughs of text and image recognition
- Discussion: Why rule-based systems fail for unstructured data

Learning Resources

- 3Blue1Brown – Neural Networks (Parts 1–2)
<https://www.youtube.com/watch?v=aircAruvnKk>
 - StatQuest – *Neural Networks Explained*
<https://www.youtube.com/watch?v=CqOfi41LfDw>
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Day 5 — Why AI Projects Fail

Teaching

- Common AI failure patterns
- No data
- Wrong problem framing
- No success metrics
- Overuse of LLMs
- Difference between demo AI and production AI

Progressive Session

- Case study analysis of failed AI systems
- Identify missing inputs, outputs, and metrics

Learning Resources

- Google Research – *Hidden Technical Debt in ML Systems*
<https://research.google/pubs/pub43146/>
 - Andrew Ng talks on AI project failures
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WEEK 2 — Thinking Like an AI Engineer

Day 6 — The AI Project Lifecycle

Teaching

1. Problem framing
2. Data assessment
3. Model selection
4. Training vs usage
5. Evaluation
6. Deployment
7. Monitoring

Progressive Session

- Map lifecycle for real-world examples (chatbots, resume screening)

Learning Resources

- Google ML Guide – Rules of Machine Learning
<https://developers.google.com/machine-learning/guides/rules-of-ml>
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Day 7 — Data: The Real Bottleneck

Teaching

- Structured vs unstructured data
- Labeling cost and effort
- Data leakage and bias
- Data quality vs model complexity

Progressive Session

- Design AI approaches when data is limited or unavailable

Learning Resources

- StatQuest – *Training Data Explained*
<https://www.youtube.com/watch?v=HQPIS1C9Nw>
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Day 8 — Rules vs ML vs LLMs

Teaching

- When rules outperform ML
- When ML outperforms LLMs
- When LLMs are the correct choice

Progressive Session

- Given multiple problem statements, choose the correct approach and justify it

Learning Resources

- Chip Huyen – *When Not to Use Deep Learning*
<https://www.youtube.com/watch?v=5O46Rqh5zHE>
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Day 9 — Communicating AI Decisions to Management

Teaching

- Translating business goals into AI language
- Setting realistic expectations
- Cost, risk, and feasibility discussions
- Saying "No" to incorrect AI requirements

Progressive Session

- Role-play exercises with management-style requirements

Learning Resources

- Andrew Ng – *AI for Everyone* (Business-focused modules)
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Day 10 — Week 1-2 Mini Capstone

Teaching

- Review of key concepts

- Preparation for next phase (Model selection & cost analysis)

Progressive Session (Assessment)

Each participant must: - Select a real-world problem - Explain: - Whether it is an AI problem - The AI category
- Required data - Risks and limitations

Outcomes by End of Week 2

Participants should be able to: - Explain AI concepts clearly without buzzwords - Identify classification vs regression problems - Decide between rules, ML, and LLMs - Ask the right questions when receiving vague requirements - Speak confidently about AI fundamentals in interviews

Teaching Guidance

- Avoid heavy mathematics
 - Avoid framework-specific coding
 - Focus on decision-making and explanation
 - Encourage verbal reasoning and justification
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This document serves as the foundation for all subsequent weeks, including model selection, cost analysis, architecture design, and hands-on AI projects.